

ARNAV VERMA

+1(551) 359-3439 • av4445@nyu.edu

linkedin.com/in/namesarnav • github.com/namesarnav • namesarnav.com



EDUCATION

New York University, Courant Institute of Mathematical Sciences – New York, NY

Expected May 2028

Master of Science, Computer Science

Coursework: **Advanced Algorithms**, Machine Learning, Deep Reinforcement Learning

University of North Texas – Denton, TX

Jan 2023 – May 2026

Bachelor of Science, Computer Science, Minor Mathematics GPA 3.70

Coursework: **Data Structures and Algorithms**, Object-Oriented Programming, Operating Systems, Natural Language Processing, Database

SKILLS

Languages	C/C++, Python, Go , SQL, Bash, JavaScript, TypeScript, HTML/CSS
ML/AI	PyTorch , TensorFlow, Hugging Face Transformers, scikit-learn, LangChain
Data	PostgreSQL , SQLite, Redis, DynamoDB, FAISS, Pinecone
Backend	FastAPI, Flask, Express, Prisma
Tools	Docker, Kubernetes , Git, GitHub Actions, Weights & Biases, MLflow
Certifications	CNCF Certified Kubernetes Application Developer (CKAD), Cred ID: LF-5ridapwt2z

EXPERIENCE

Undergraduate Researcher – University of North Texas

May 2025 – May 2026

- Fine-tuned and evaluated 42 model configurations across dense and MoE architectures on temporal and causal reasoning tasks using a novel Hungarian algorithm-based metric across 7+ generalization datasets
- Collaborated with department researchers and led systematic analysis of inductive, deductive, and abductive reasoning strategies, finding that inductive prompting improved cross-dimensional consistency by **40%** over CoT combinations
- Developed adversarial and domain-shift evaluation datasets exposing **60%+** performance drops, and fine-tuned RoBERTa, T5, and LLaMA baselines for comparison with prompted LLMs

PUBLICATIONS

- Iqbal, F., Das, S., Dutt, R., **Verma, A.**, Ray Choudhury, S. "Evaluating Multi-Dimensional Generalization of Large Language Models in Temporal Extraction Tasks." **Accepted to Findings of AACL-IJCNLP 2026**
- **Verma, A.**, Iqbal, F., Ray Choudhury, S. "Do LLMs Reason Causally, or Just Pattern Match? A Multi-Strategy Evaluation." Under Review

PROJECTS

Semantic Video Search Engine – Python, FAISS, CLIP, SQLite

[Code](#) | [Demo](#)

- Engineered a CLIP-based video search engine over **205 videos and 4,415 frames**, building the backend engineering pipeline in Python and a React-based frontend interface returning ranked, playable timestamps from plain language queries
- Made a query path encoding text with CLIP, ranking 4,415 normalized vectors by FAISS inner product, then joining SQLite metadata
- Produced scene-cut sampling on Bhattacharyya distance in HSV histograms, threshold at 0.35 inside a measured **0.20–0.65** plateau
- Proved scene-aware sampling raises **Recall@5 from 85.7% to 100%** on a purpose-built cut-dense benchmark

Synapse – Go, PostgreSQL, Redis, React, TypeScript, Docker, OpenTelemetry

[Code](#) | [Demo](#)

- Built Synapse, **a distributed workflow automation tool** with fencing tokens and transactional state advancement, sustaining **443 tasks/s** on 4 cores with zero failures across **6,000** executions at 2× capacity and **17/30/44 ms p50/p95/p99 latency at 100 exec/s**
- Engineered a PostgreSQL-backed execution queue and event log with multi-process workers and no message broker
- Designed a persisted real-time event pipeline with bounded WebSocket fan-out and optional Redis acceleration, delivering **90K** events with zero loss and **3–5 ms p50** push latency across **100** concurrent subscribers
- Created a from-scratch expression parser/evaluator, SSRF-protected HTTP nodes, **AES-256-GCM** secrets, node-level replay, and a React Flow DAG editor; shipped with Docker, Prometheus, OpenTelemetry, and 205 race-tested Go tests at **79.9%** coverage

Marigold – Python, TypeScript, Pandas, Scikit-learn, PostgreSQL, Redis

[Code](#) | [Demo](#)

- Authored a PDF to Quiz study platform serving **500 users**, using Flask, Postgres, Redis and React, generating **15 quizzes and flashcards** per upload through multiple LLM providers
- Deployed a **SAKT machine learning** knowledge tracing model at **0.756 held-out AUC** across **277,720** interactions from **3,022 students**, preprocessing data with Pandas in Jupyter and reproducing the training run end to end
- Containerized the ML pipeline and application with Docker to streamline deployment, reducing environment setup time from 20 minutes to under 1 minute for new contributors